

Data Mining

Practical Session 1

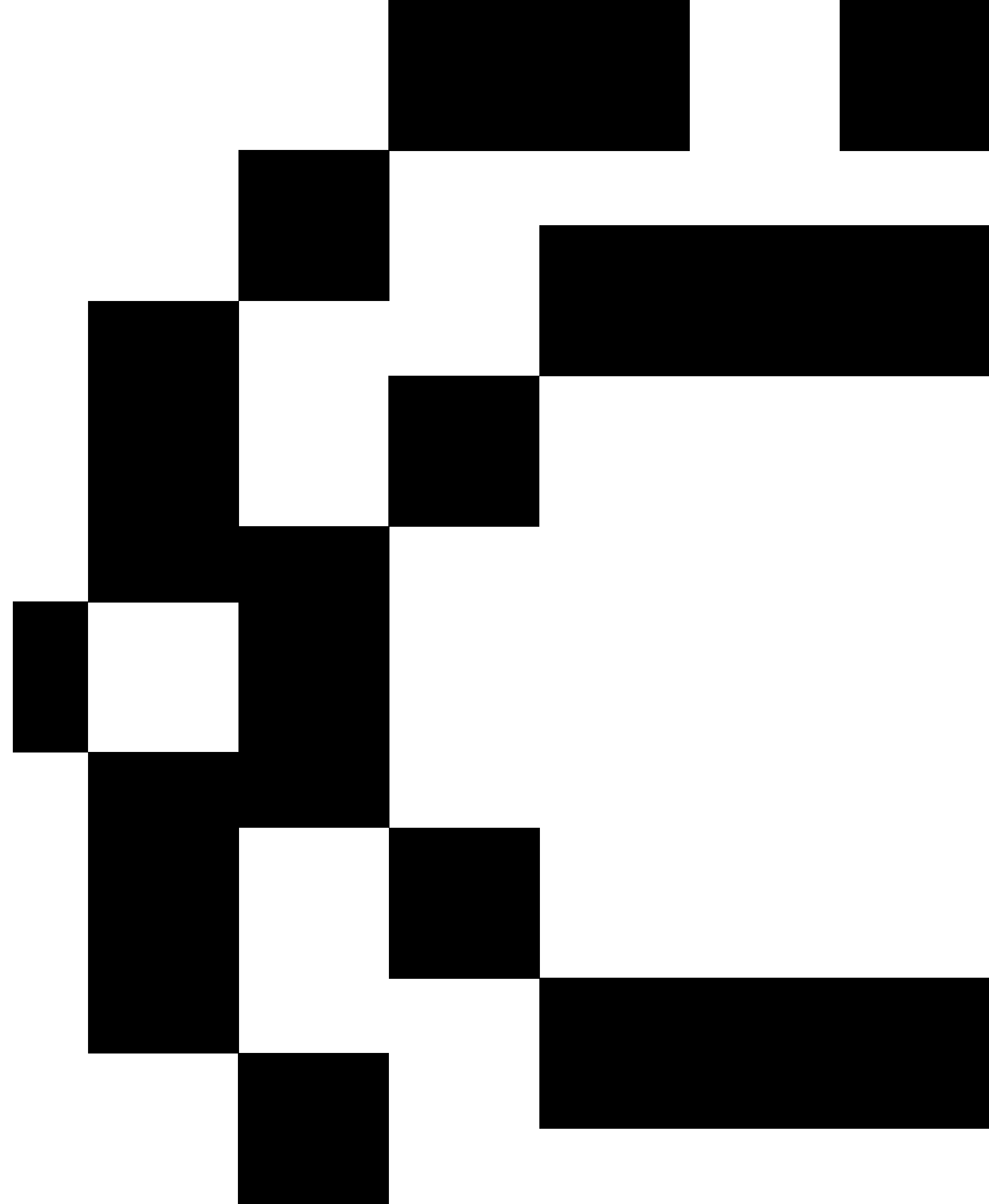
Fall Semester 2026-2027
Master's in Data Science and Advanced Analytics

Ana Pedro Caleiro

acaleiro@novaims.unl.pt

Yehor Malakhov

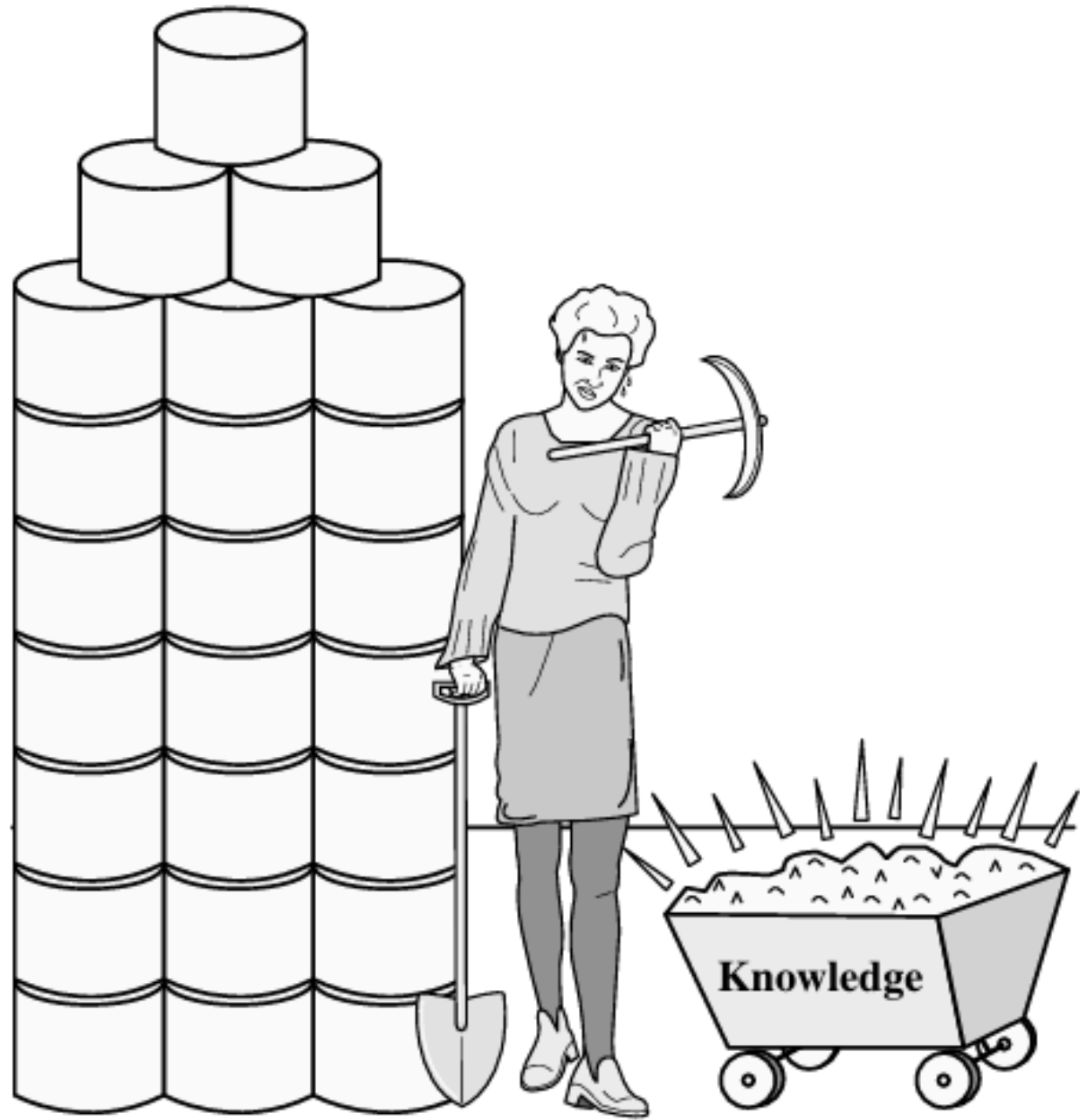
ymalakhov@novaims.unl.pt



Class Topics

- **Data Mining Context**
- Where to find the Resources
- Project Overview
- Project Methodology & Structure
- General Policies & Guidelines
- Success Strategies
- Questions & Support

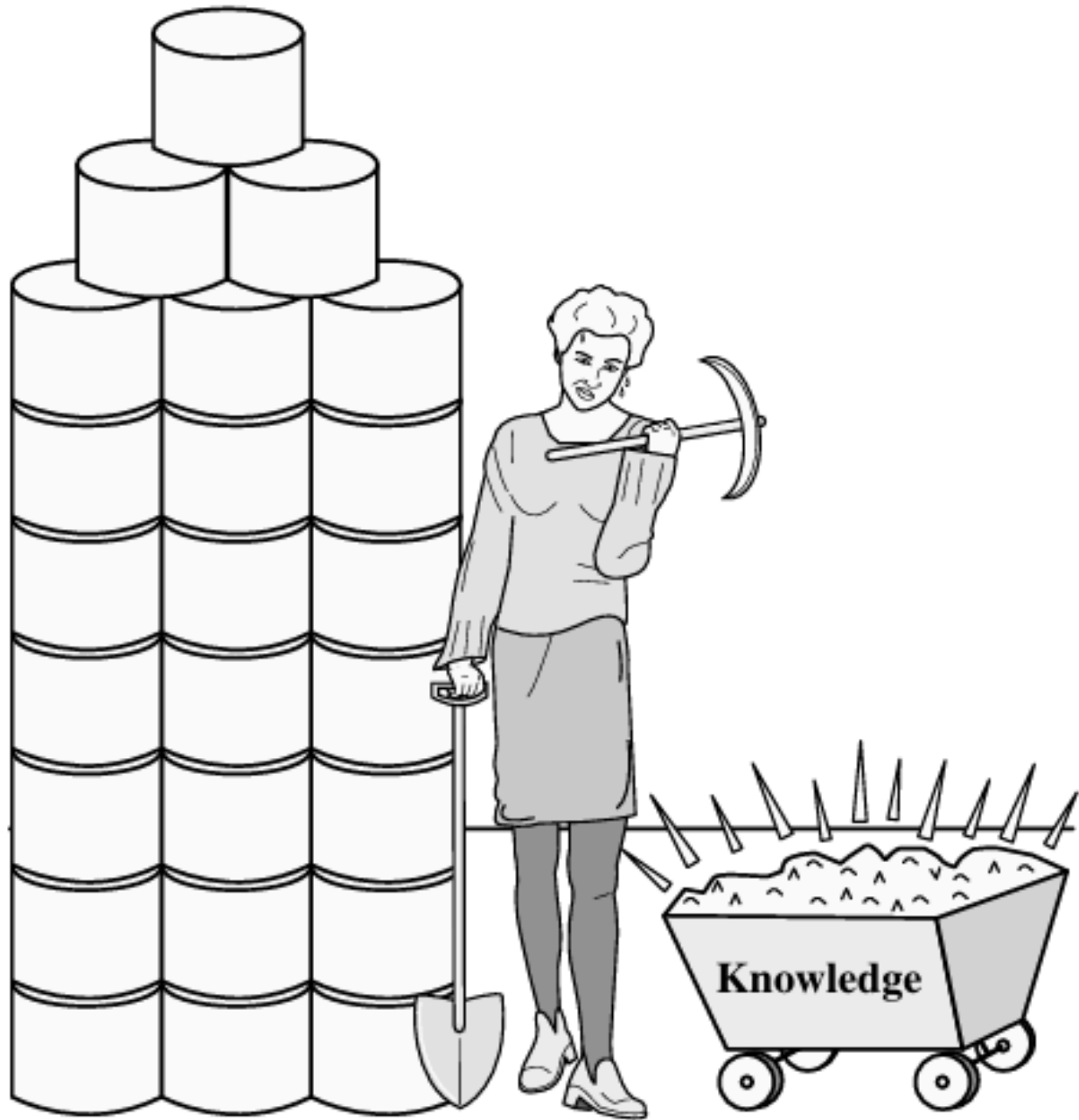
What is Data Mining?

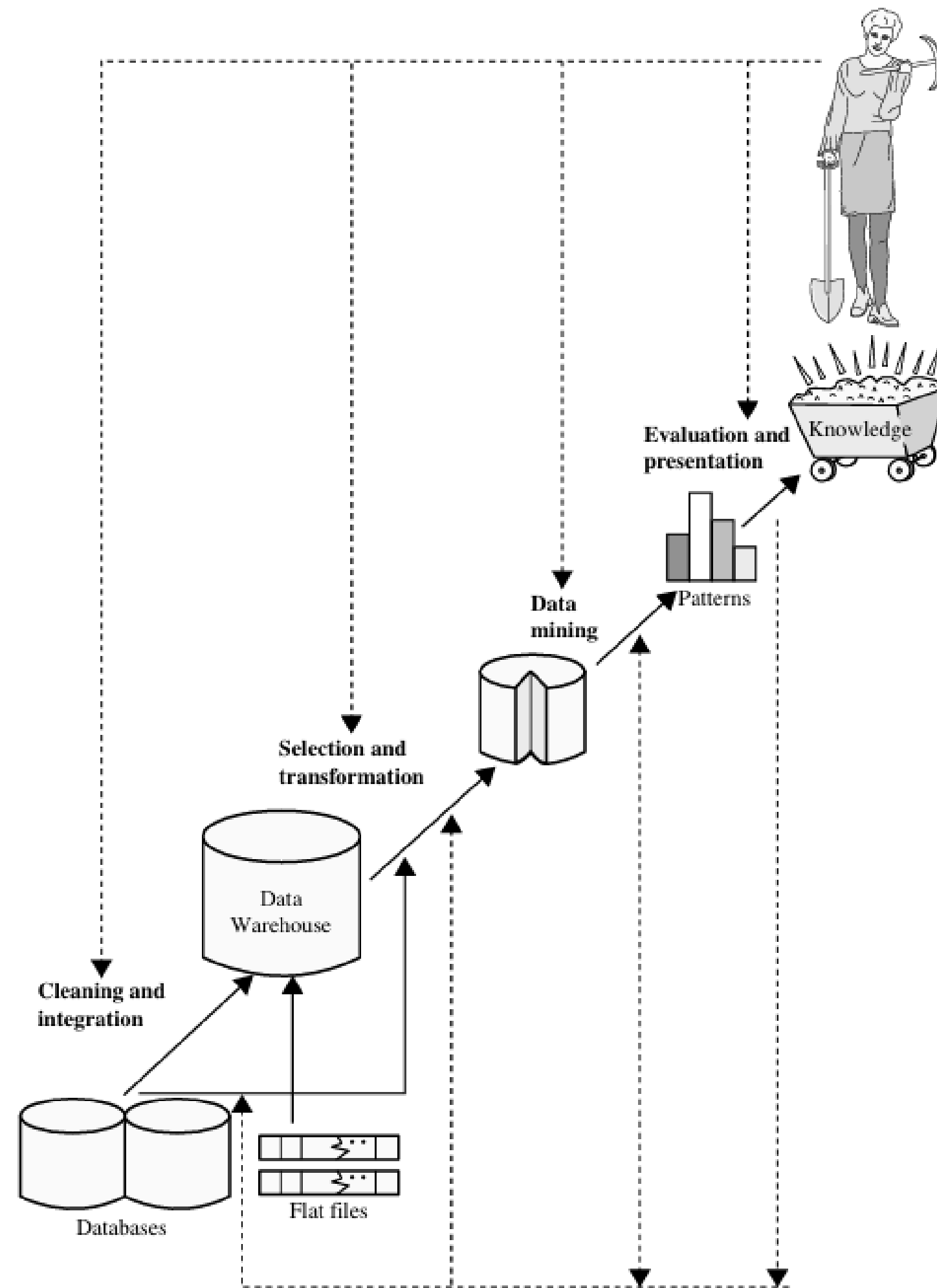


[Data Mining Intro](#)

What is Data Mining?

Data mining is the **process of discovering interesting patterns and knowledge from large amounts of data**. The data sources can include databases, data warehouses, the Web, other information repositories, or data that are streamed into the system dynamically.



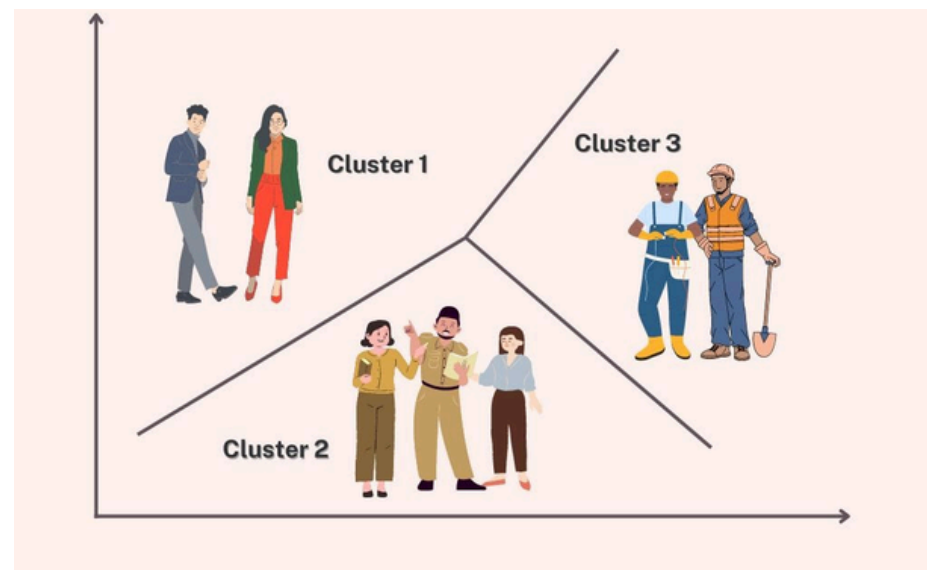


Knowledge discovery process

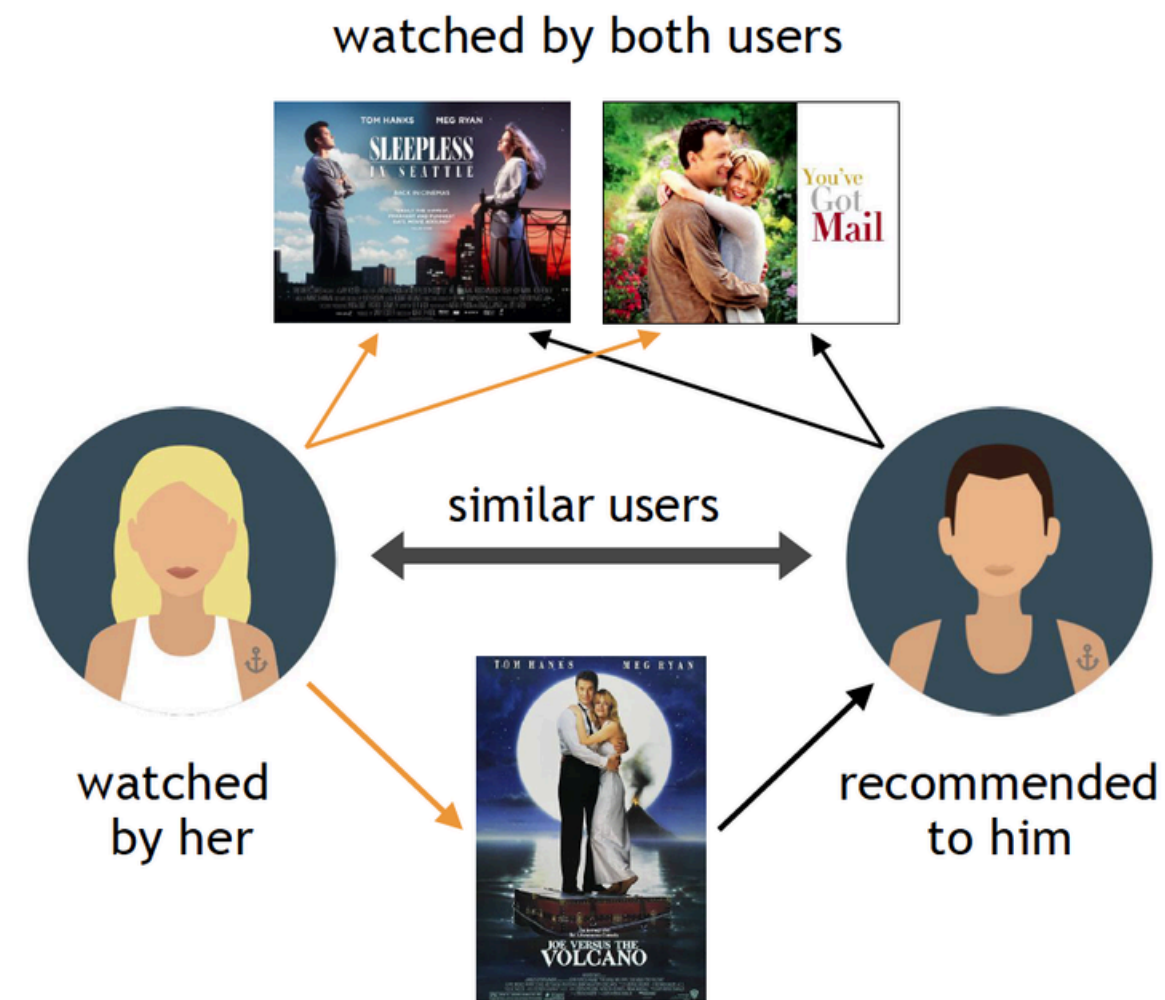
1. **Data cleaning:** to remove noise and inconsistent data
2. **Data integration:** where multiple data sources may be combined
3. **Data selection:** where data relevant to the analysis task are retrieved from the database
4. **Data transformation:** where data are transformed and consolidated into forms appropriate for mining by performing summary or aggregation operations
5. **Data mining:** an essential process where intelligent methods are applied to extract data patterns
6. **Pattern evaluation:** to identify the truly interesting patterns representing knowledge based on interestingness measures
7. **Knowledge presentation:** where visualization and knowledge representation techniques are used to showcase the findings

Data mining in the real world...

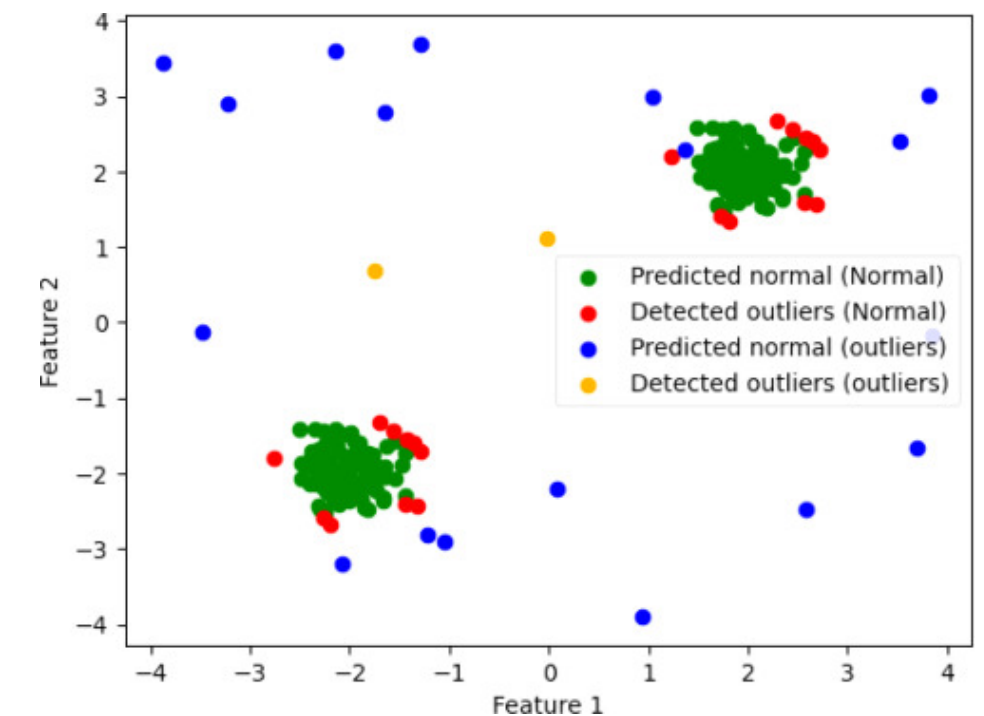
Customer segmentation



Recommendation systems



Anomaly detection



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Where to find the Resources?

Let's go to moodle: <https://elearning.novaims.unl.pt/course/view.php?id=5261>

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/ Outono | Fall

202627 - Data Mining - Turma TP1 | TP2 - S1

[Disciplina](#) [Configurações](#) [Participantes](#) [Pauta](#) [Atividades](#) [Mais ▾](#)

▼ [Geral](#) [Contrair tudo](#)

 [Anúncios](#)

▼ [Nova subsecção](#)

 [Project Guidelines](#)

Project information and data

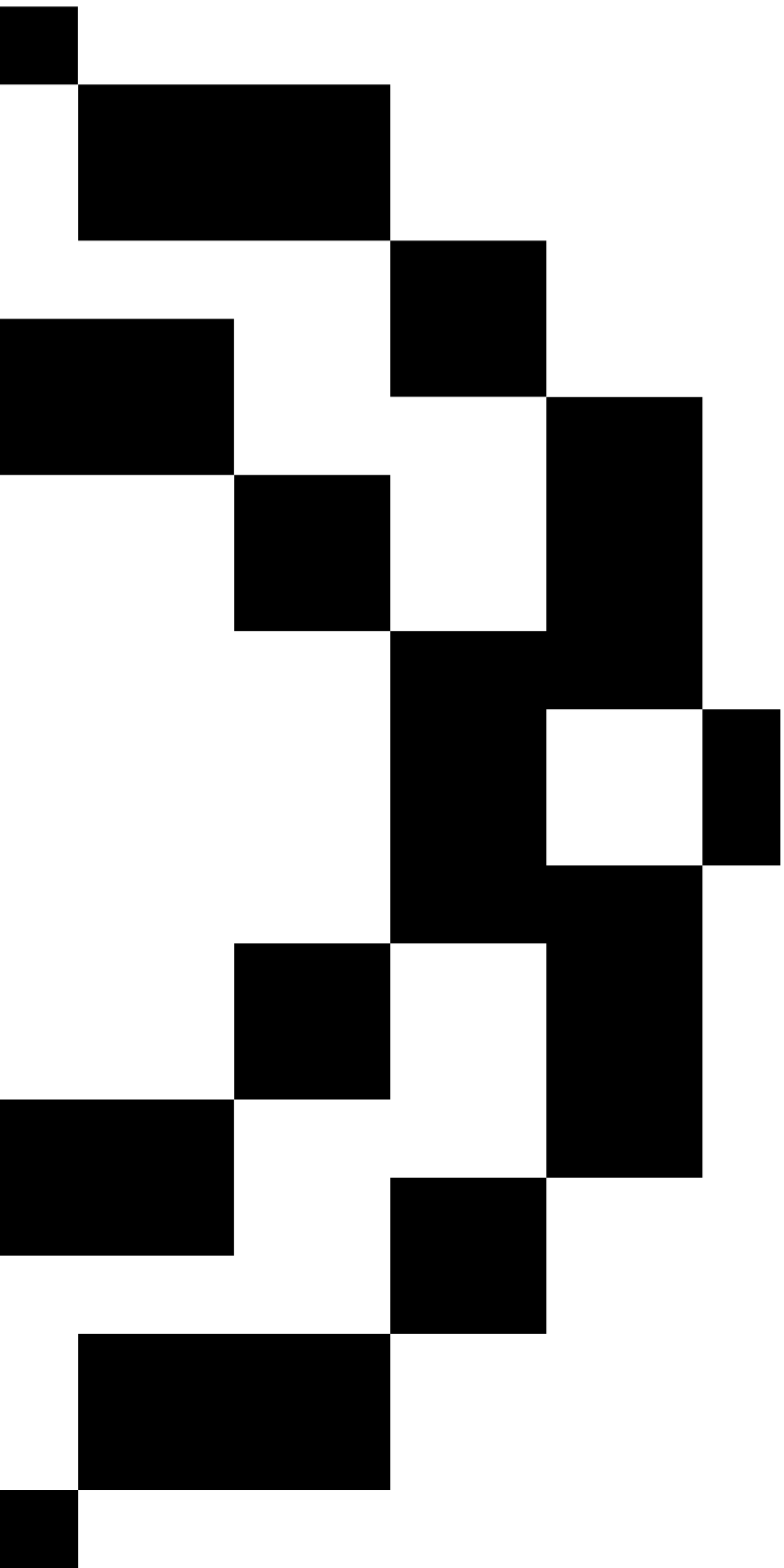
 [Project Group Membership](#)

 [Practical Sessions](#)

Practical
classes link
(Github)

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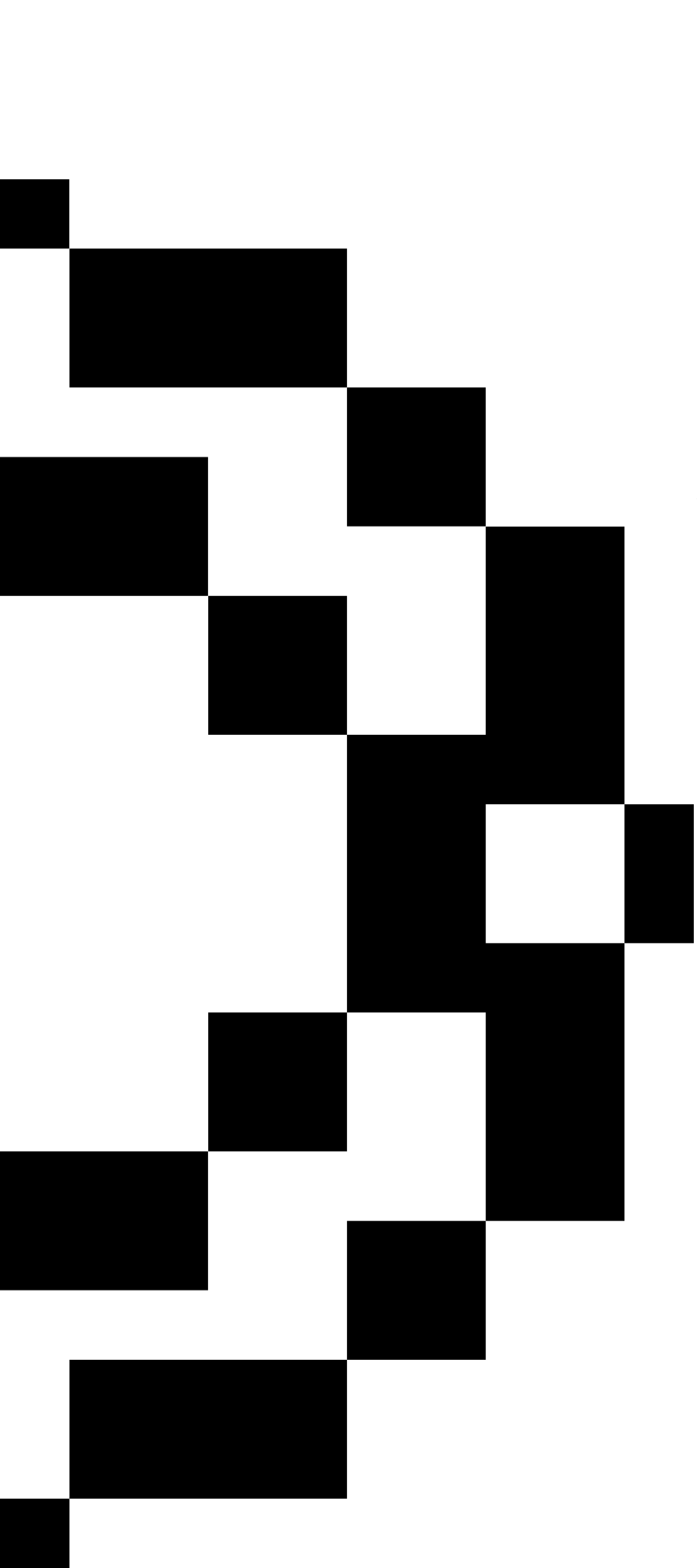


Project at a Glance

- **Client:** Amazing Creative Desk Coworking (ACDC)
- **Objective:** Customer segmentation for personalized marketing
- **Duration:** Full semester project with 3 deliverables (including discussion)
- **Team Size:** 3-4 students per group
- **Total Points:** 100 points (20 + 60 + 20)
- **Methodology:** CRISP-DM framework

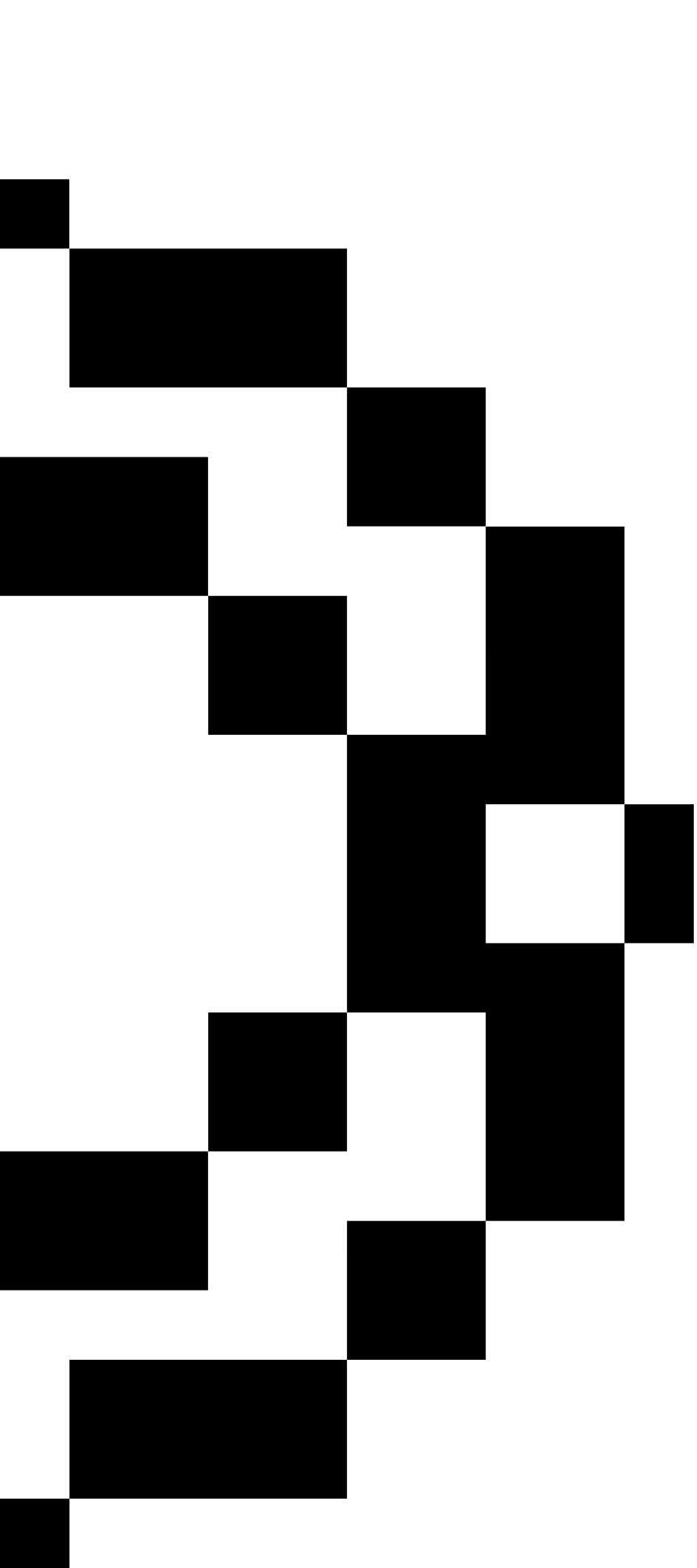
Deliverables (Project 45% & Exam 55%)

Deliverable	Components	Points	Focus
EDA	Notebook + Slides + Discussion	20	Data Understanding
Clustering	Notebook + Report	60	Modeling & Strategy
Discussion	Group discussion	20	Individual assessment



Amazing Creative Desk Coworking (ACDC)

- **Challenge:** Design personalized services and marketing strategies
- **Context:** Highly competitive coworking industry
- **Opportunity:** Leverage data-driven customer segmentation
- **Dataset:** 3-year membership and space-usage data

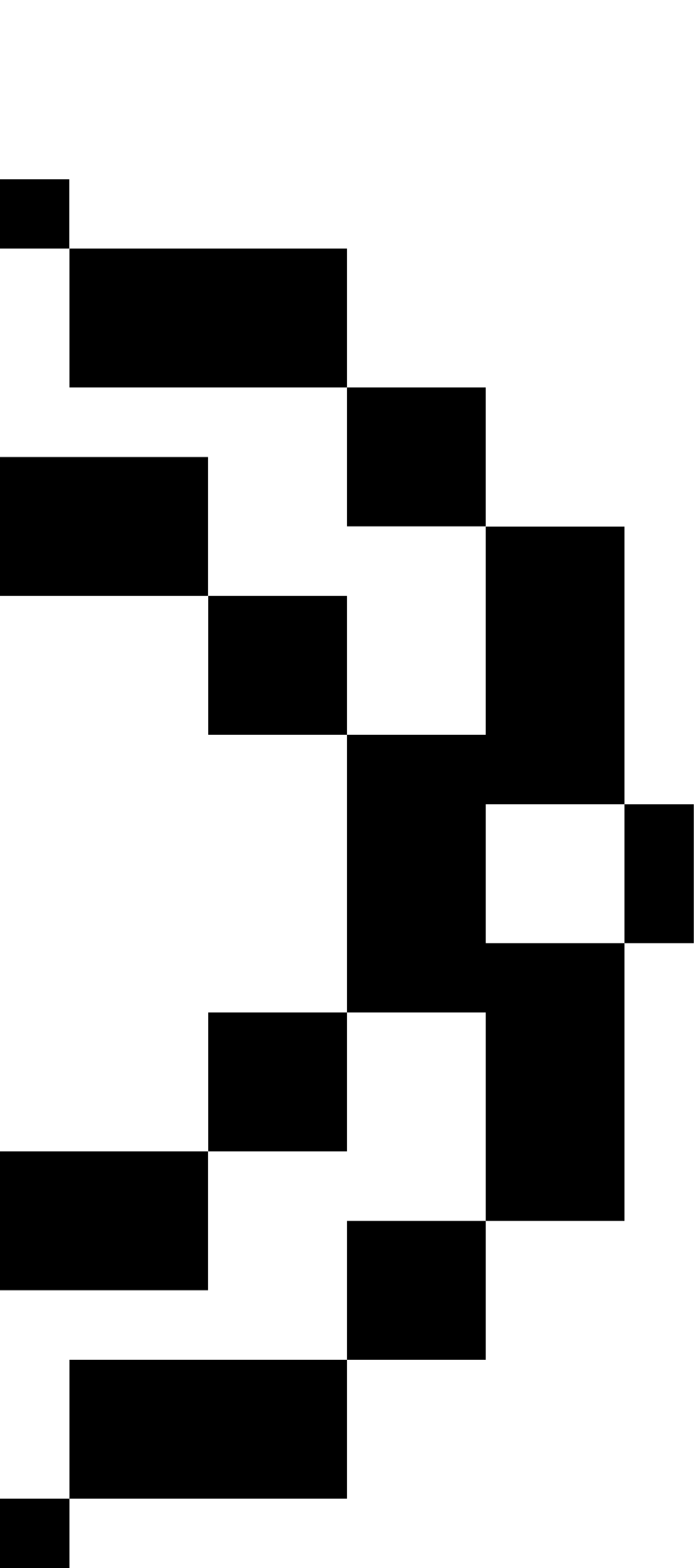


Customer Segmentation Perspectives

- **Value-based segmentation:** Economic contribution grouping
- **Behavioral segmentation:** Usage habits and space-attendance behaviors
- **Demographic segmentation:** Age and attribute patterns
- **Goal:** Integrate perspectives into a comprehensive marketing strategy

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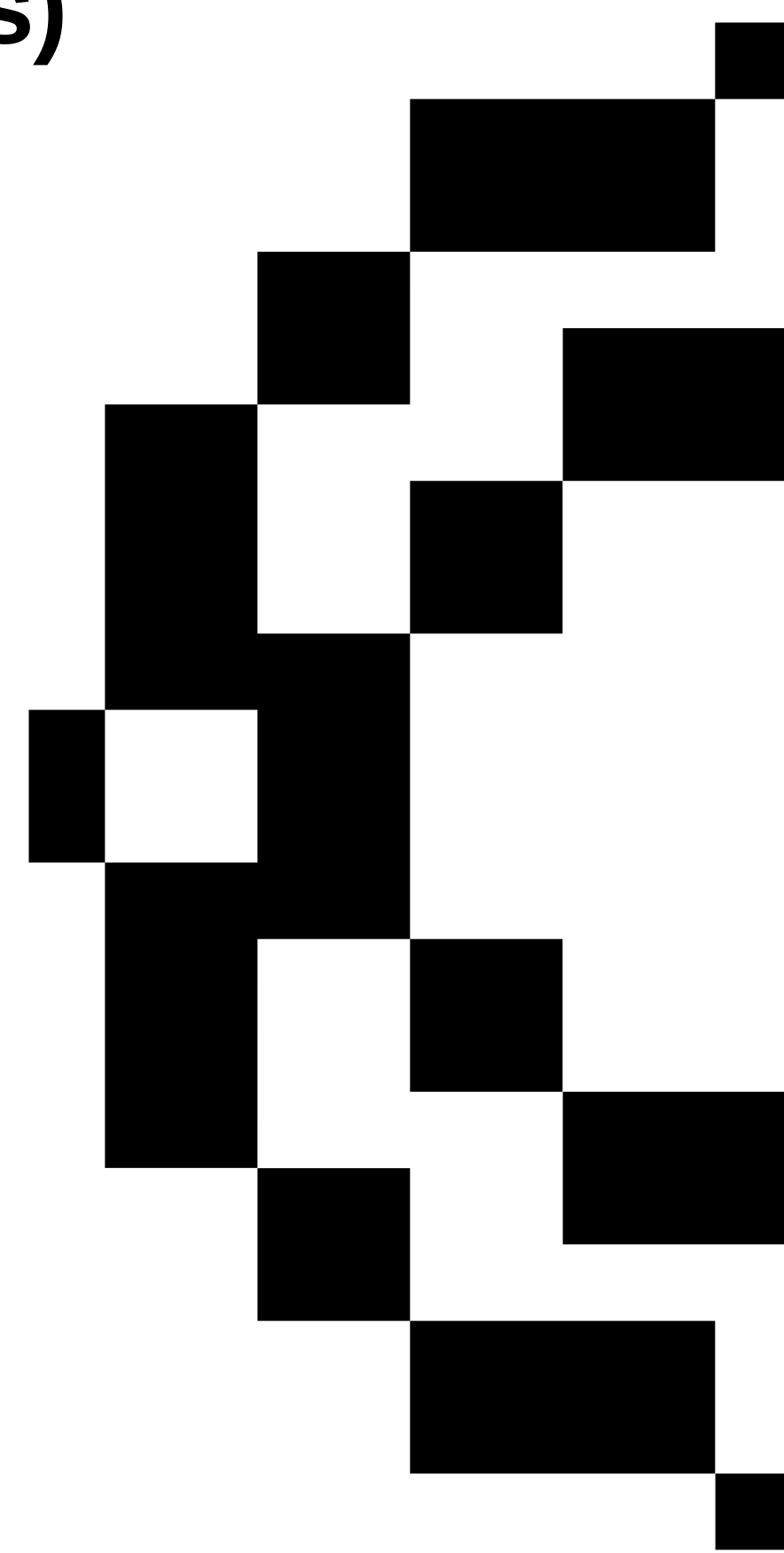
CRISP-DM Methodology

- **Business Understanding:** Industry challenges and segmentation needs
- **Data Understanding:** Dataset exploration and quality assessment
- **Data Preparation:** Feature engineering and preprocessing
- **Modeling:** Multi-perspective clustering analysis
- **Evaluation:** Validation and business interpretation
- **Deployment:** Strategic recommendations and implementation

Deliverable 1: Exploratory Data Analysis (20 points)

Key Tasks

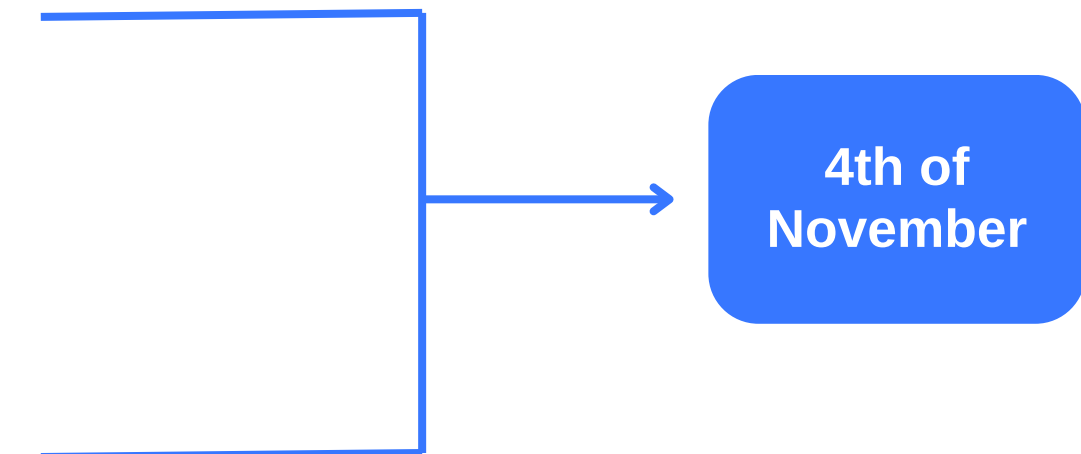
- Conduct descriptive statistics and visualizations
- Assess data quality issues and clustering reliability
- Identify preliminary behavioral signals
- Develop and justify engineered features



Deliverable 1: Exploratory Data Analysis (20 points)

Components Overview

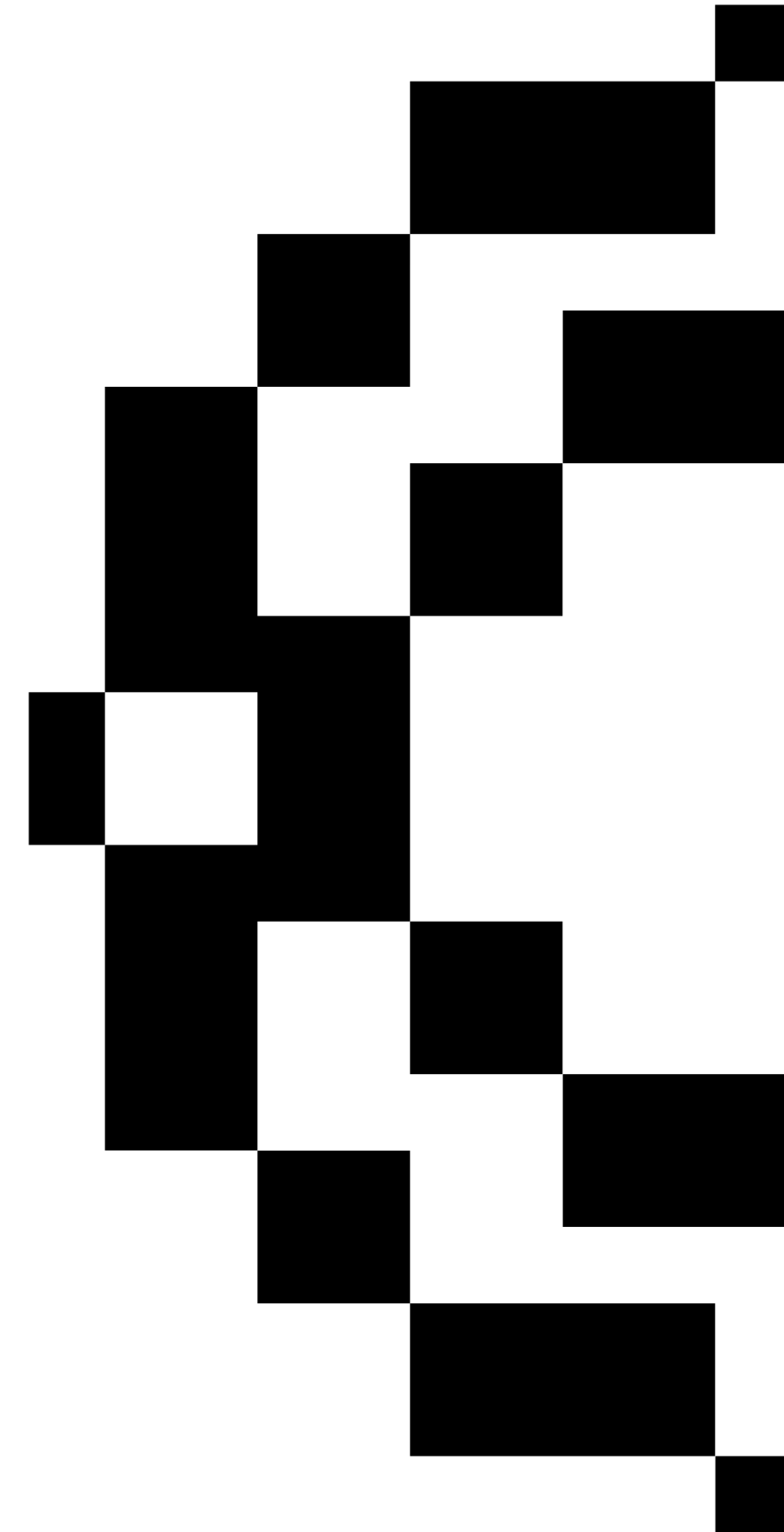
- **Component 1: Jupyter Notebook (5 points)**
 - File: `GroupXX_EDA_Code.ipynb`
 - Clean, documented code with a systematic workflow
- **Component 2: Slides (15 points)**
 - File: `GroupXX_EDA_Slides.pdf`
 - Minimum 3, maximum 5 slides of content, excluding the cover slide
- **Component 3: Group Discussion**



Deliverable 2: Clustering Analysis (60 points)

Key Tasks

- Prepare dataset for modeling
- Apply multiple clustering perspectives
- Compare at least two clustering approaches per perspective
- Propose final merged segmentation solution
- Integrate insights into coherent framework



Deliverable 2: Clustering Analysis (60 points)

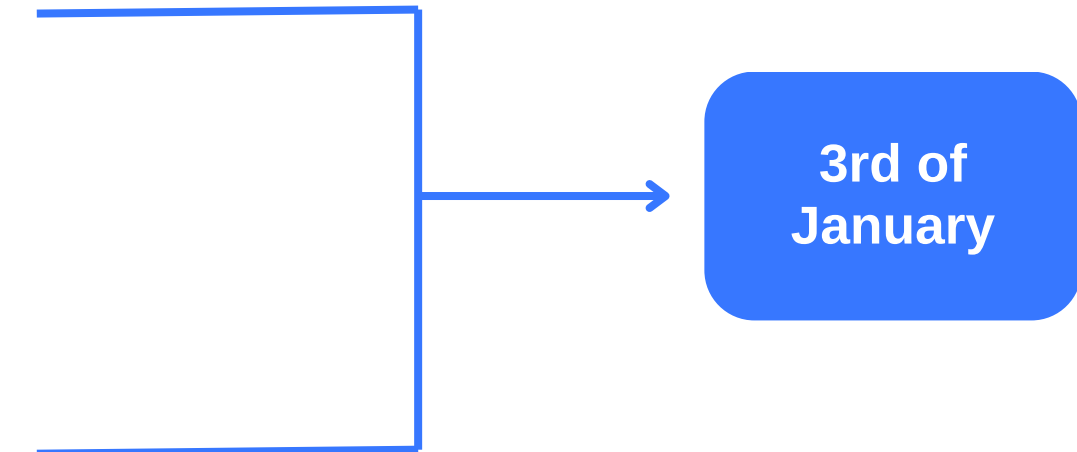
Components Overview

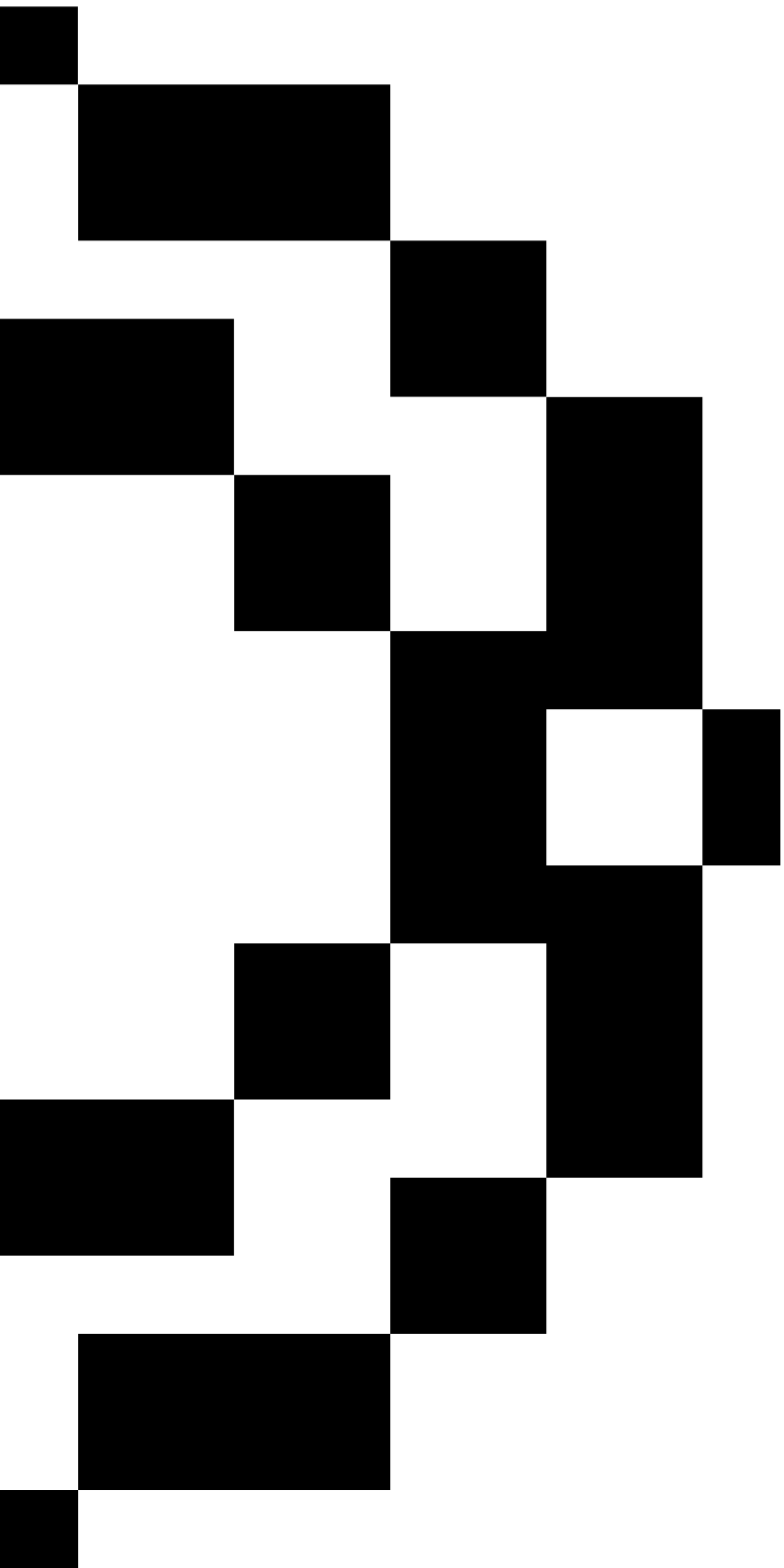
- **Component 1: Jupyter Notebook (5 points)**

- File: `GroupXX_Clustering_Code.ipynb`
- Multiple algorithms implemented and compared

- **Component 2: Report (55 points)**

- File: `GroupXX_Clustering_Report.pdf`
- Maximum 10 pages with strategic recommendations





Bonus Components (Optional)

Deliverable 1 Bonus Options

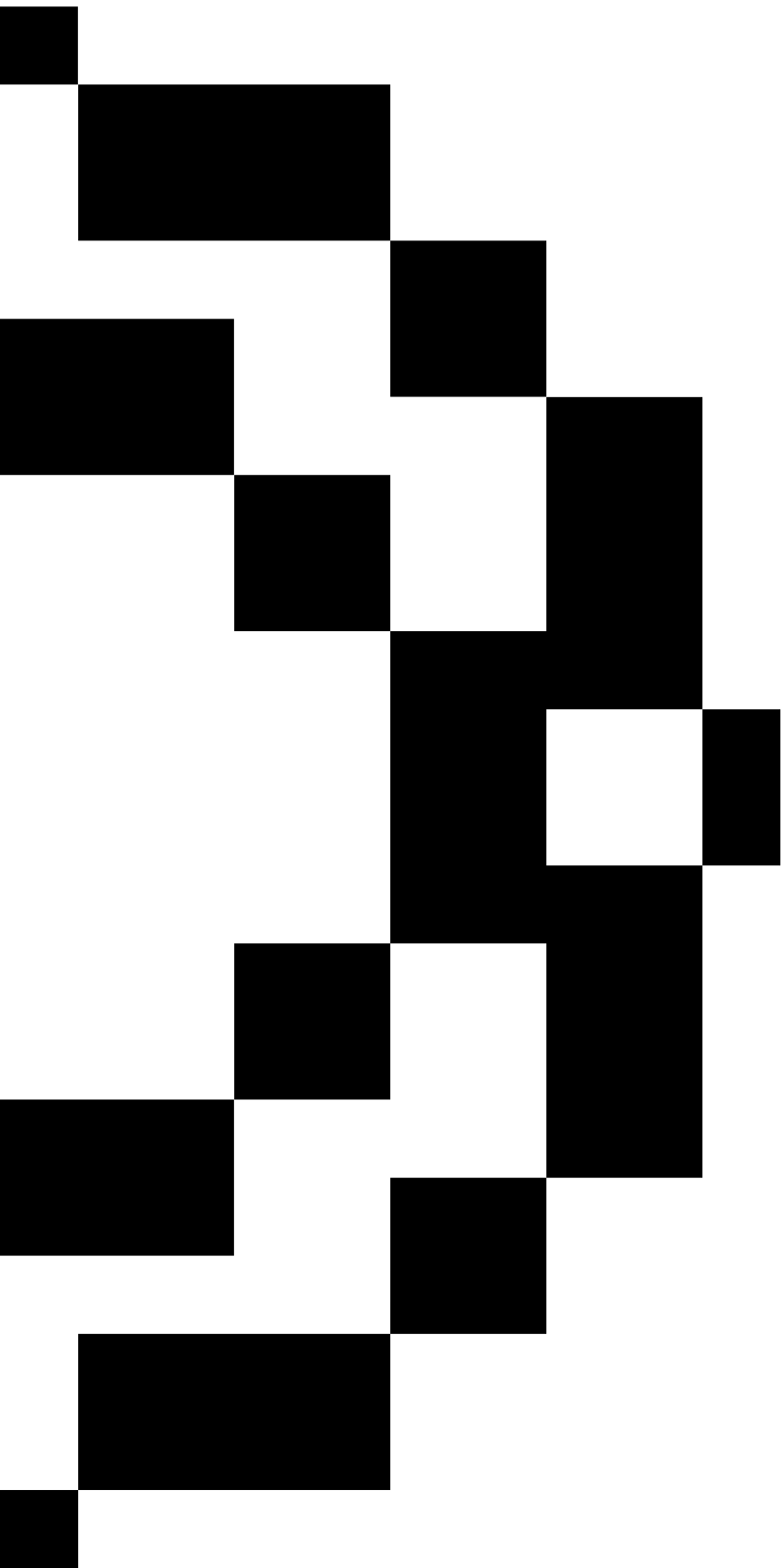
Up to 20% additional credit

Deliverable 2 Bonus Options

Up to 20% additional credit

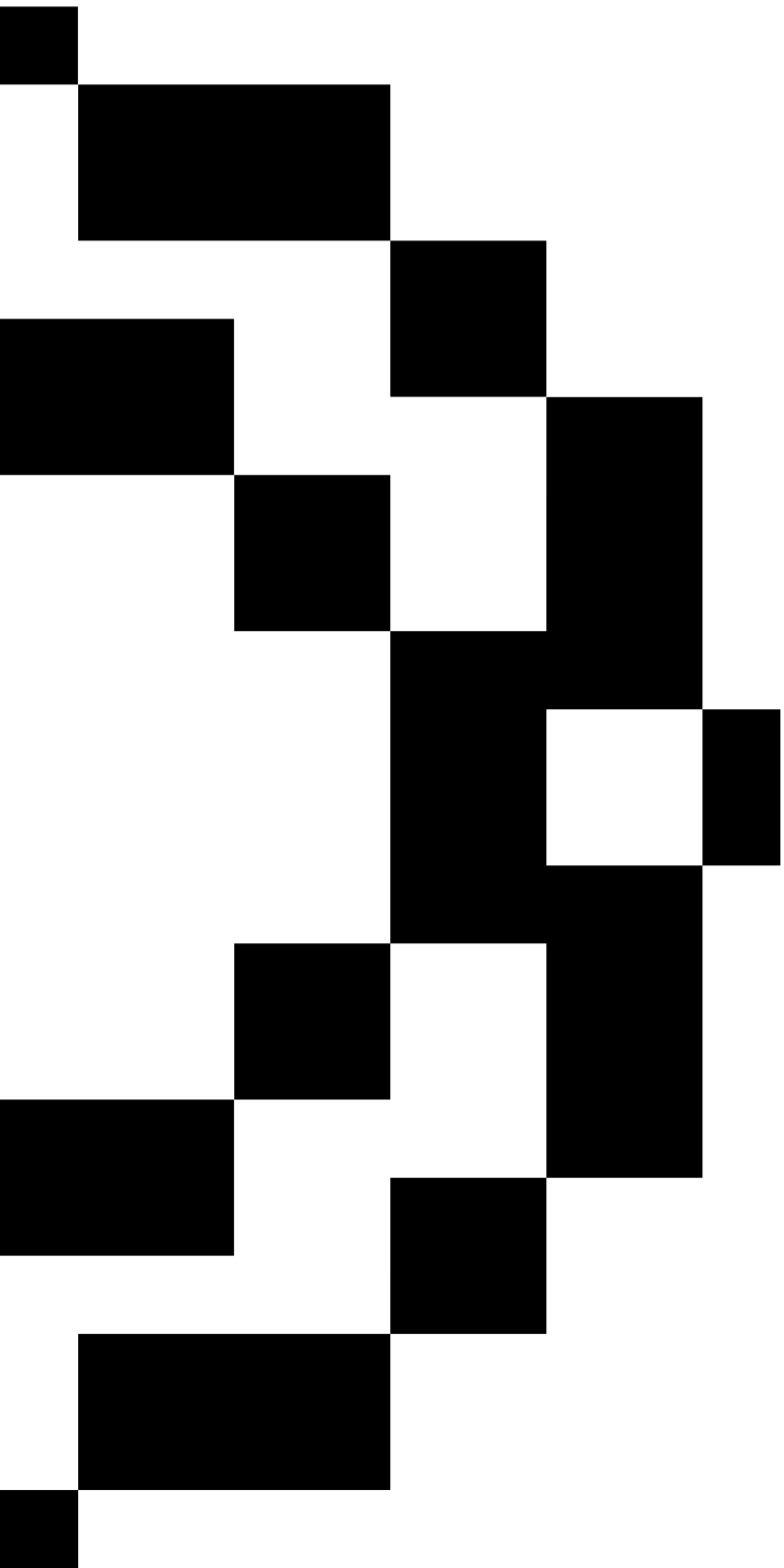
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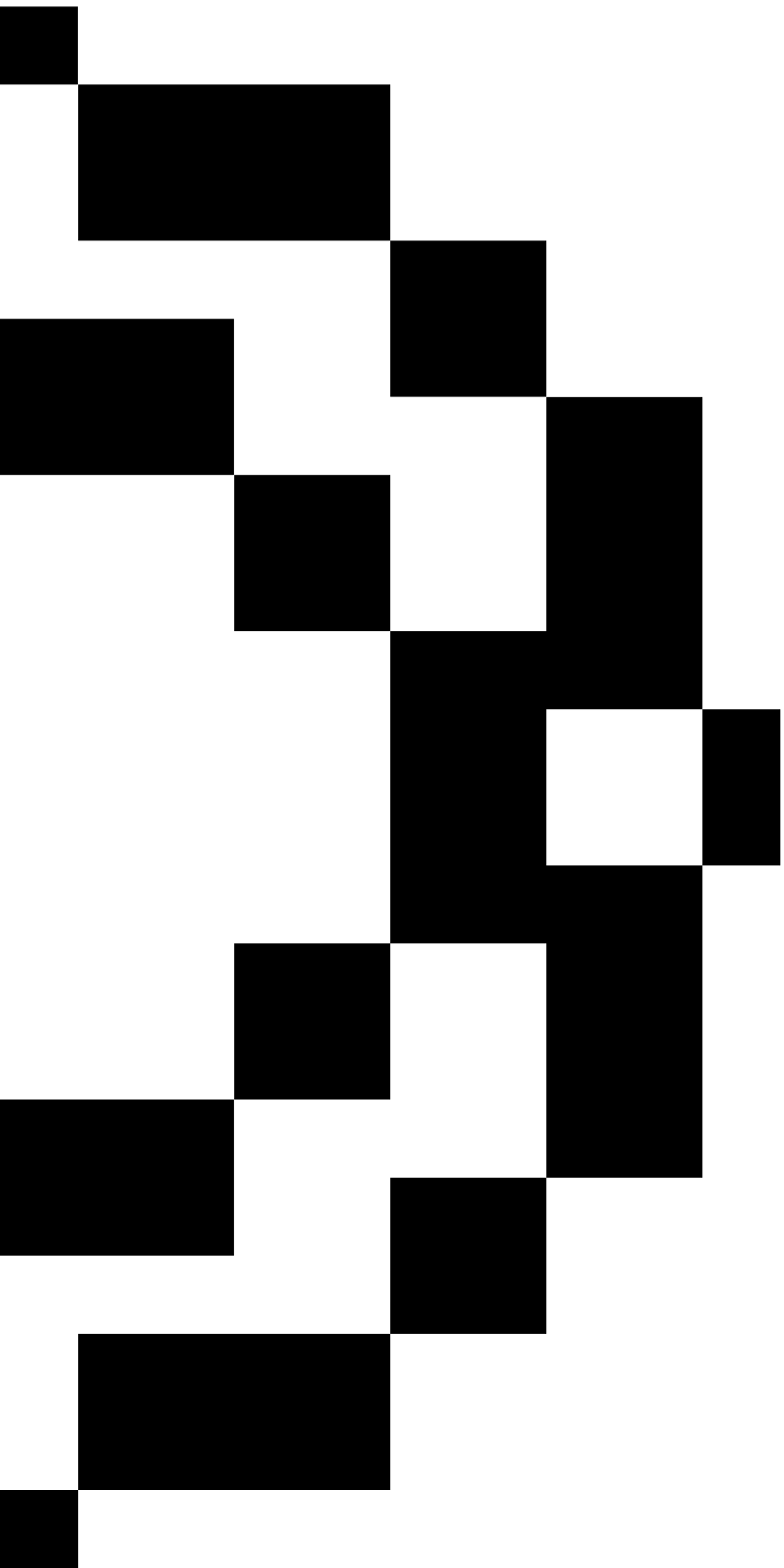
Group Composition

- **Maximum:** 4 students per group (recommended: 3)
- **Enrollment:** All students must register on Moodle
- **Deadline:** Group formation before first delivery
- **Changes:** Not recommended after first delivery



AI Tools Usage

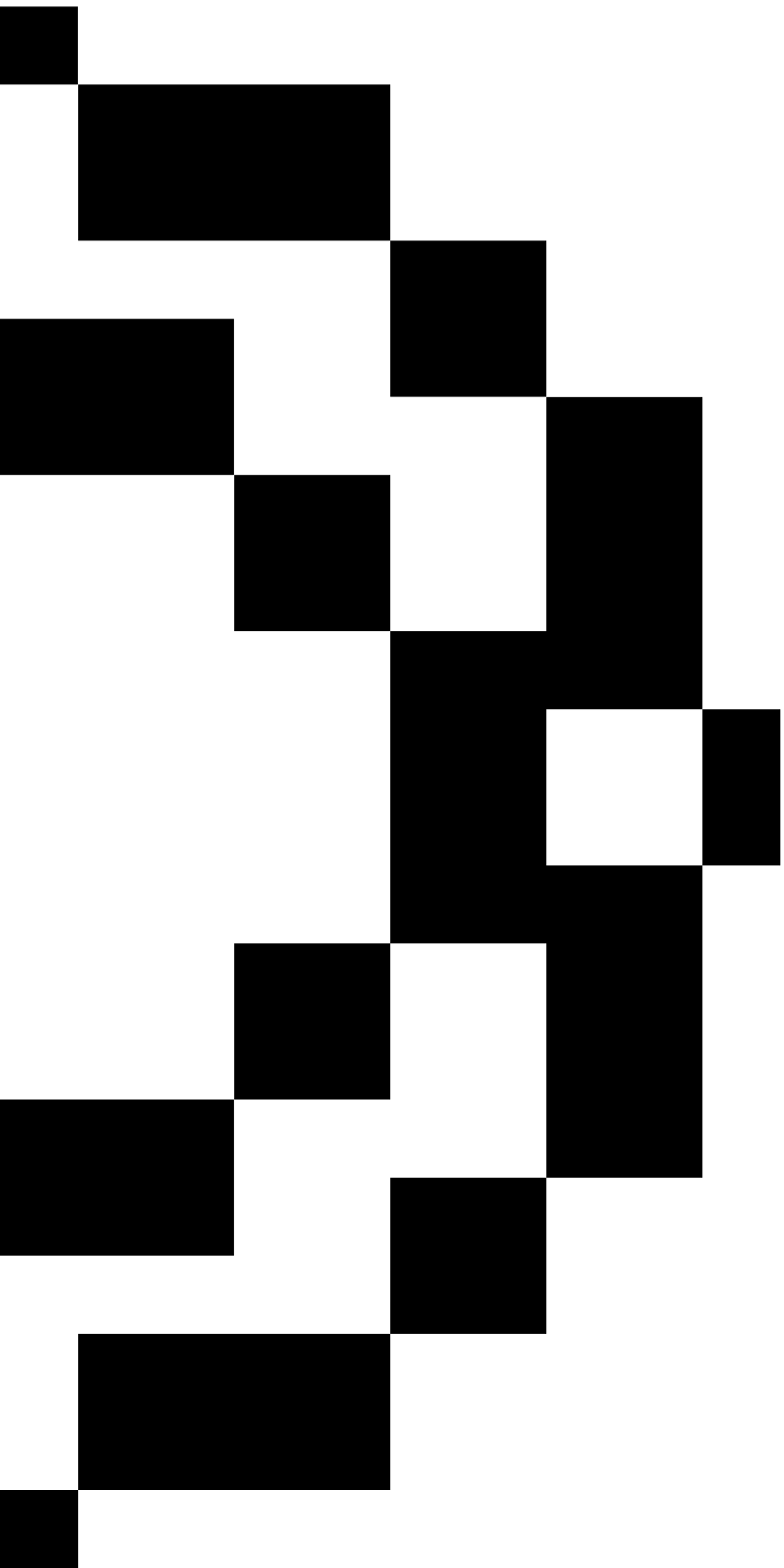
- **Permitted:** With full disclosure required
- **Documentation:** AI Usage Statement in annexes
- **Responsibility:** Students accountable for all content
- **Requirement:** Original analysis must exceed AI contribution
- **Explanation:** Students may be asked to explain AI-generated content



Mandatory Annexes

All reports must include:

- **AI Usage Statement:** Specific tools and usage documentation
- **Contribution Statement:** Individual student contributions
- **Responsibility Statement:** Group accountability certification

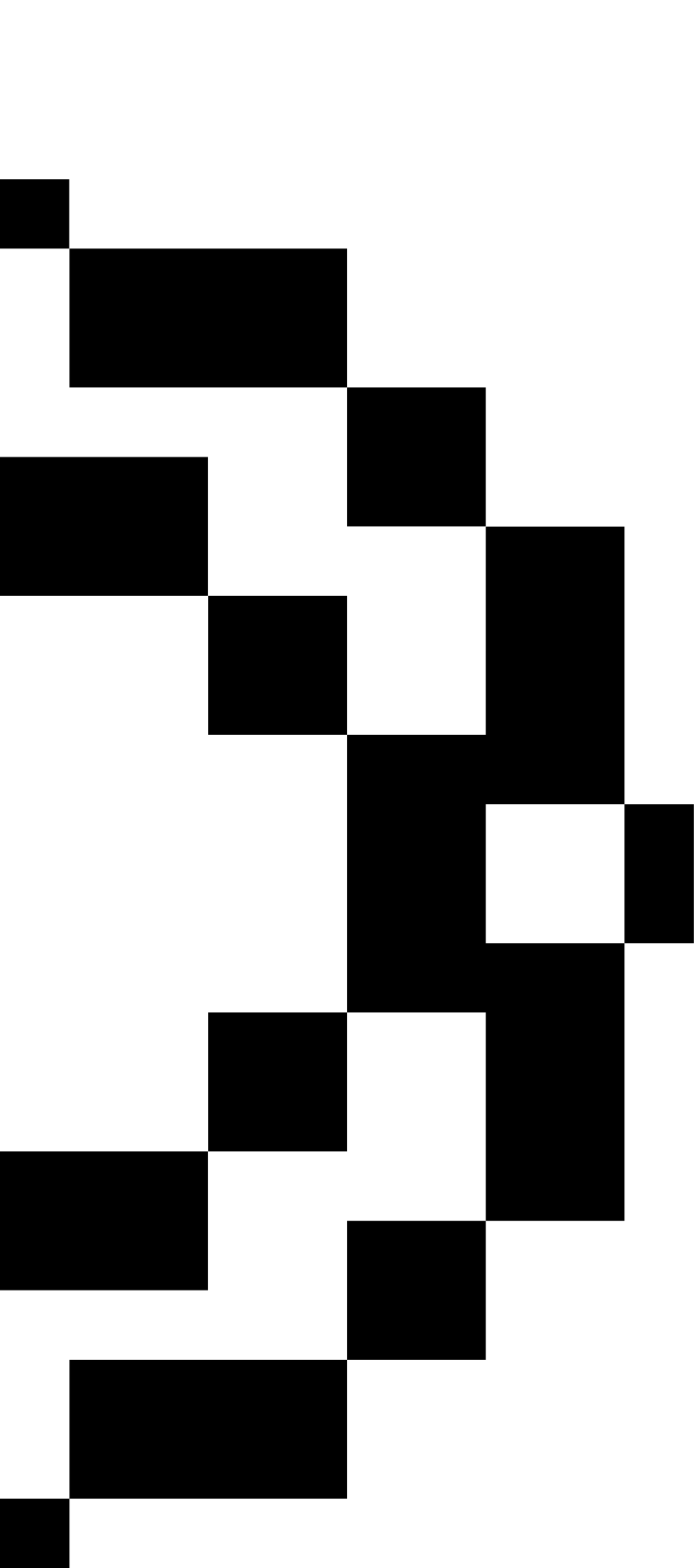


Penalty System

- **Late Delivery:** 10% penalty per day
- **Guideline Deviations:** Additional deductions may apply
- **Quality Standards:** All components must meet passing standards
- **Plagiarism Check:** All reports undergo plagiarism verification

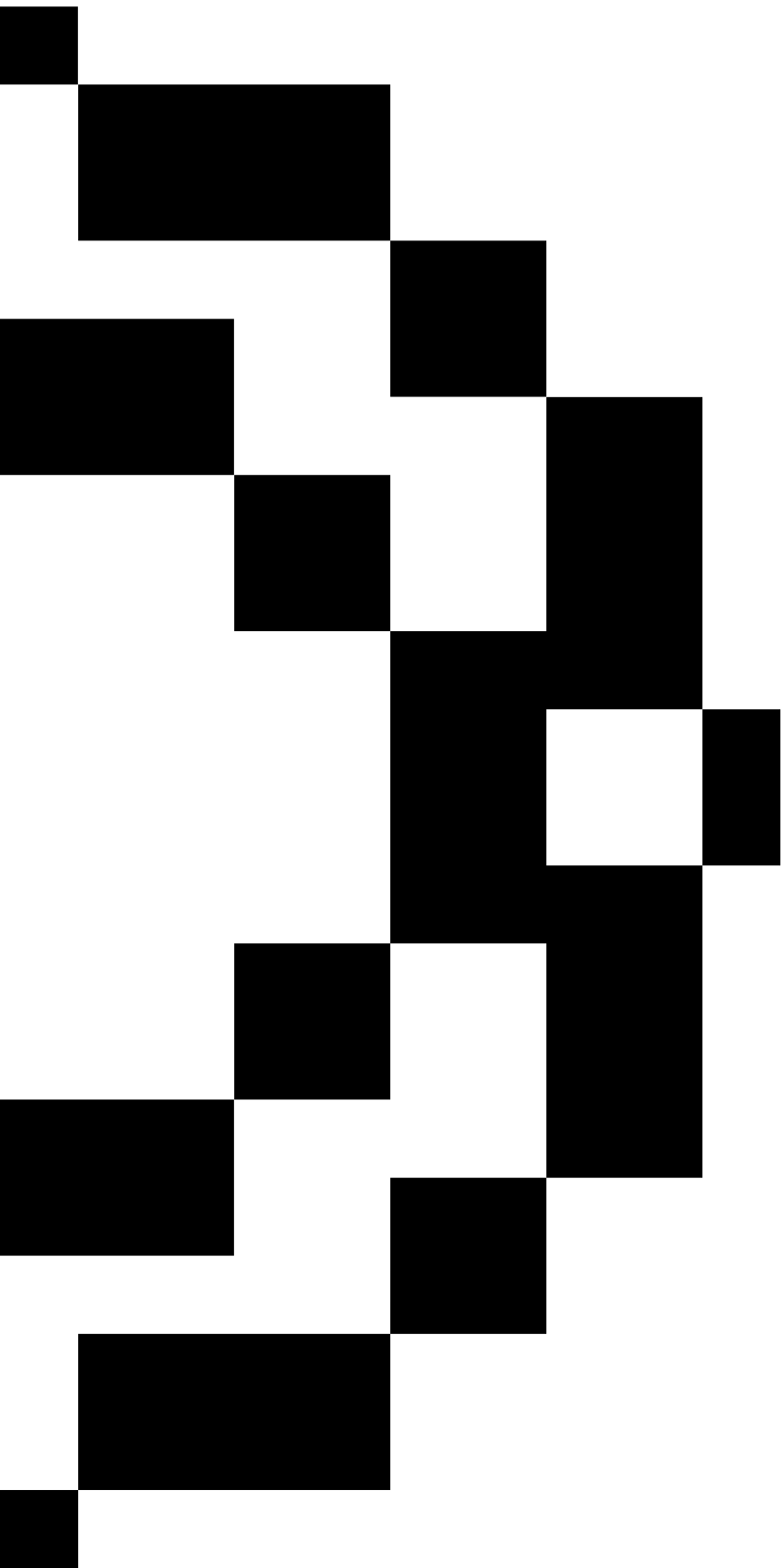
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Report

- **Focus:** Insights and interpretations, not just descriptions
- **Structure:** Follow provided frameworks strictly
- **Business Context:** Connect technical findings to strategic value
- **Evidence-Based:** Support all claims with data analysis
- **Executive Communication:** Accessible to non-technical stakeholders

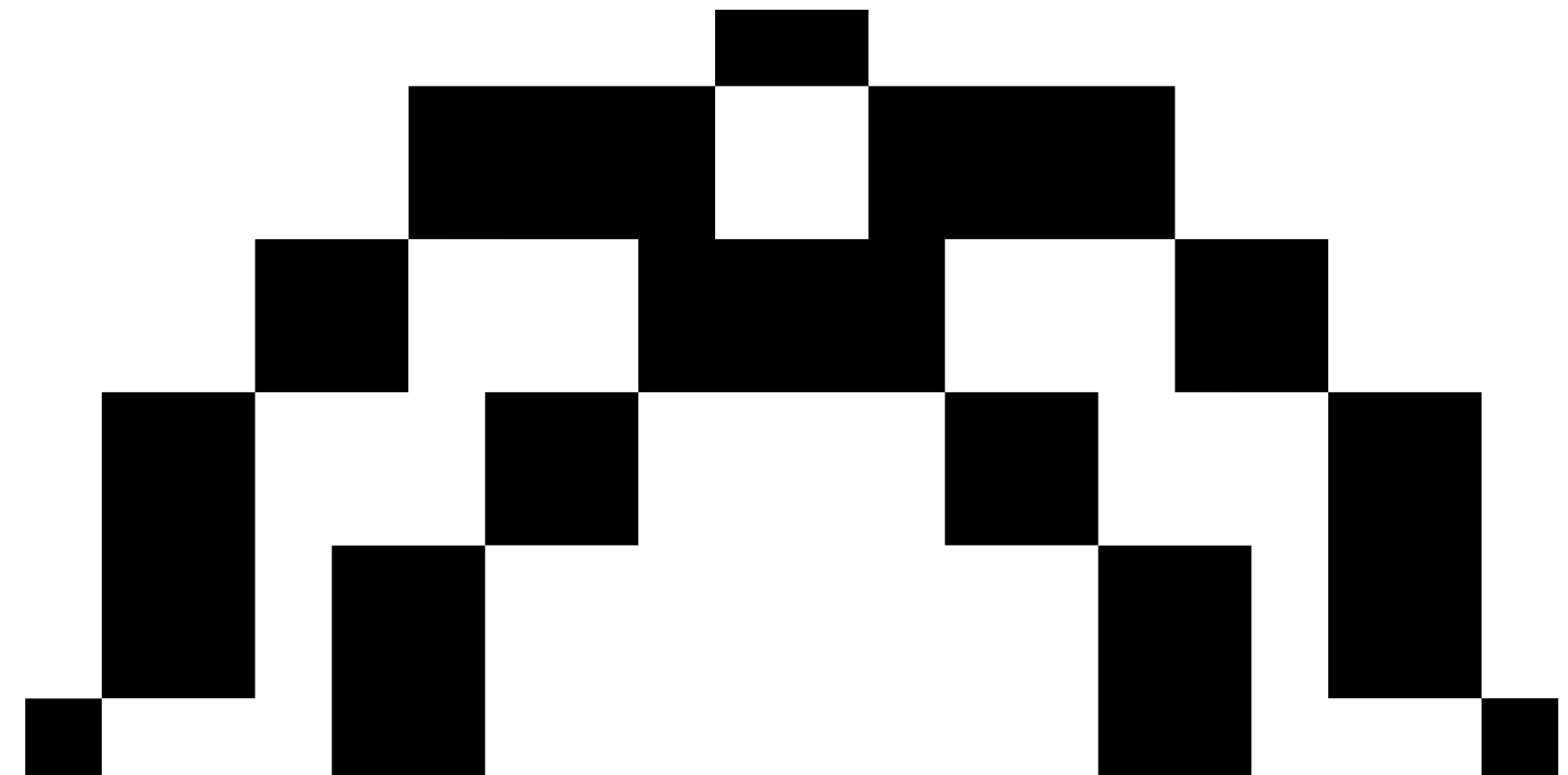


Technical Implementation

- **Code Quality:** Clean, documented, reproducible workflows
- **Multiple Methods:** Compare different clustering approaches
- **Validation:** Use appropriate metrics and statistical testing
- **Integration:** Merge perspectives into coherent solution
- **Business Translation:** Connect technical results to actionable strategies

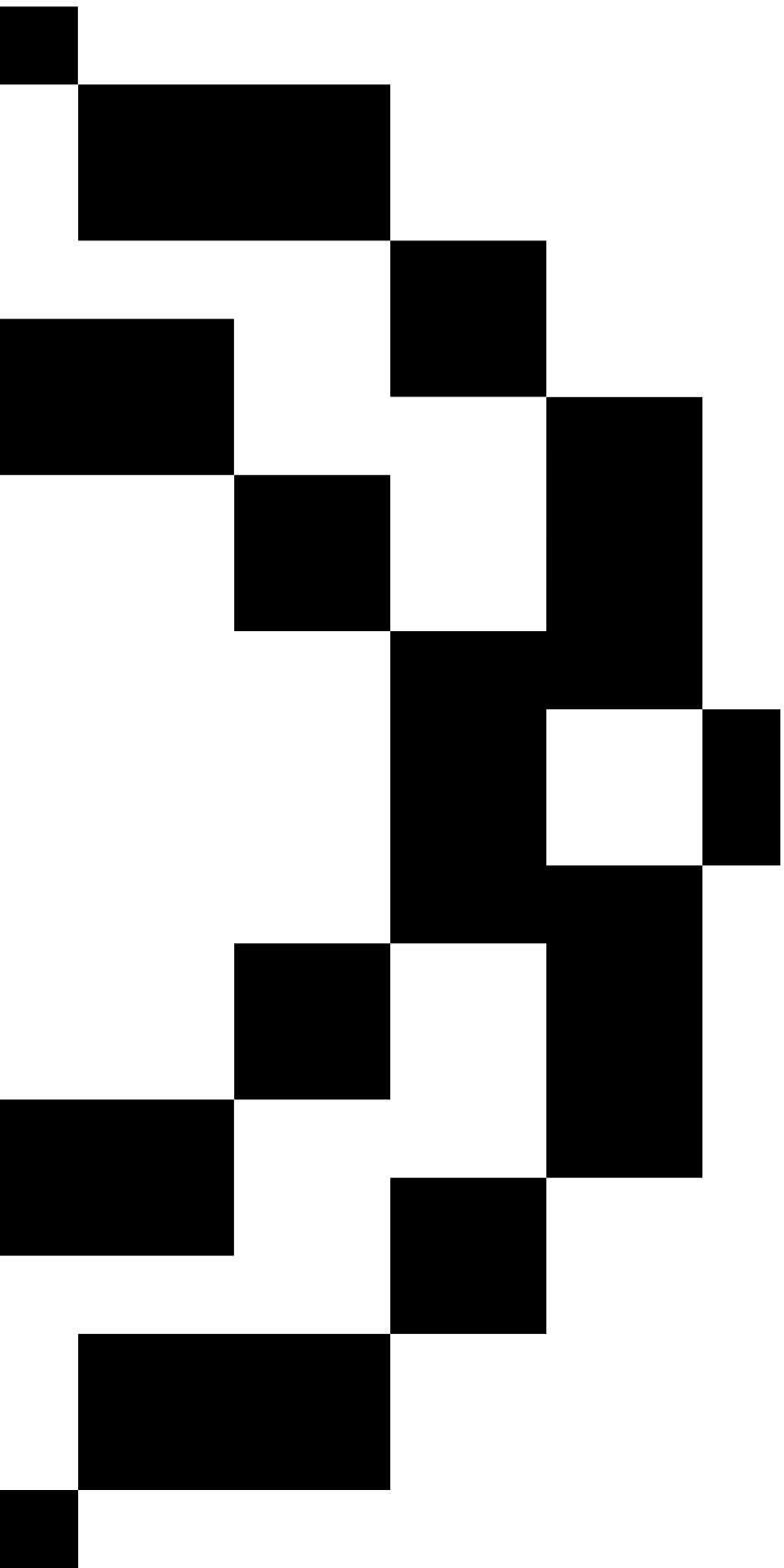
Final Reminders

- **Originality:** Ensure all work represents your group's analysis
- **Documentation:** Maintain clear records of all processes
- **Timeline:** Plan deliverables well ahead of deadlines
- **Quality:** Prioritize depth over breadth in analysis
- **Business Value:** Always connect findings to ACDC strategic needs



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Getting Help

- Course materials and documentation
- Office hours and TA support
- Moodle discussion forums
- Group collaboration and peer learning

Obrigada!

Acreditações e Certificações da NOVA IMS



Cofinanciado por

